WORKED SOLUTION

STEP I 1998, Question 14

Probability and Statistics • Geometric Probability

STEP I 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q14

The key idea

Everything about where the coin lands is captured by where its *centre* lands. By the regular pattern of the board, we may take the centre to be uniformly distributed over a single 2×2 square, and then each outcome corresponds to a region of that square. Turning "crosses a line" and "covers an intersection" into conditions on the centre's distances from the lines and corners turns the probability into an area, which is a quadratic in a that we can maximise.

Solution

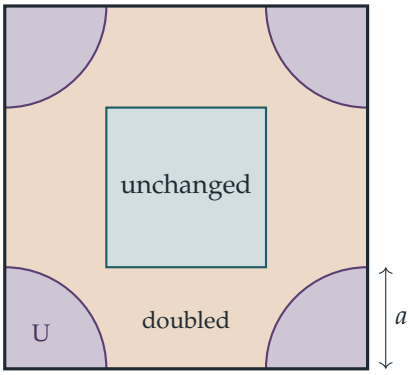
The probability that the tax is doubled

Model the coin's centre as landing uniformly at random in one square of the grid; the board is large, so edge effects are negligible, and every square looks the same. Put coordinates on this square so that it is $[0,2] \times [0,2]$, with the grid lines along its edges.

Let U be the distance from the centre to the nearest vertical line and V the distance to the nearest horizontal line. Since the x-coordinate is uniform on $[0,2]$, $U = \min(x, 2-x)$ is uniform on $[0,1]$; similarly V is uniform on $[0,1]$, and U, V are independent because x and y are. So (U,V) is uniform on the unit square $[0,1]^2$, and probabilities are just areas in that square.

- The coin crosses no line exactly when $U > a$ and $V > a$. This has probability $(1-a)^2$.
- The coin covers an intersection exactly when the nearest corner of the square is within a of the centre, that is, when $U^2 + V^2 < a^2$. Since $a < 1$, this quarter-disc lies wholly inside $[0,1]^2$, so the probability is $\frac{1}{4}\pi a^2$.

These two events are disjoint: if the coin covers an intersection, it certainly crosses the lines through it. (Boundary cases such as the coin just touching a line have probability zero, so it doesn't matter whether we use $<$ or $\leq$.)



The figure shows the possible positions of the centre in one square: teal (no line crossed) and plum (an intersection covered, marked U at one corner) leave the tax unchanged; amber doubles it.

So the probability that the tax is doubled is

$$\begin{aligned} p(a) &= 1 - (1-a)^2 - \tfrac{1}{4}\pi a^2 \\ &= 2a - \left(1 + \frac{\pi}{4}\right) a^2. \end{aligned}$$

Maximising the tax

To raise most tax we want the doubling to be as likely as possible, so we maximise $p(a)$ over $0 < a < 1$. It is a quadratic with negative leading coefficient, and

$$p'(a) = 2 - 2\left(1 + \frac{\pi}{4}\right) a = 0 \iff a = \left(1 + \frac{\pi}{4}\right)^{-1}.$$

This value is $\frac{4}{4+\pi} \approx 0.560$, which does lie in $(0,1)$, and since $p''(a) < 0$ it gives the maximum, as required. Writing a^* for this value, we have $(1 + \frac{\pi}{4})a^{*2} = a^*$, so the maximum value is

$$p_{\max} = 2a^* - a^* = a^* = \frac{4}{4+\pi}.$$

The expected tax after n years

? **Ask yourself** Why is the expected growth factor each year $1 + p$ and not $2p$?

In a year when the tax is not doubled it stays as it was, so it is multiplied by 1, not by 0. Let M_k be the factor for year k: $M_k = 2$ with probability p and $M_k = 1$ with probability $1 - p$. Then

$$\mathrm{E}(M_k) = 2p + (1-p) = 1 + p = 1 + \frac{4}{4+\pi} = \frac{8+\pi}{4+\pi}.$$

Assume each year's throw is independent of the others. The tax after n years is $T_n = M_1 M_2 \cdots M_n$ Slihings per yak (starting from 1), and the expectation of a product

of independent random variables is the product of their expectations, so

$$\mathrm{E}(T_n) = \mathrm{E}(M_1)\,\mathrm{E}(M_2)\cdots\mathrm{E}(M_n) = \left(\frac{8+\pi}{4+\pi}\right)^n,$$

as required.

Checks

- At $a \to 0$ the coin almost never touches a line, and at $a = 1$ it always crosses a line, so $p(1) = 1 - \frac{\pi}{4} \approx 0.215$, the chance of avoiding the corner discs. Both agree with the formula, and the maximum lies between them.
- A simulation with 400 000 random centres gave $p \approx 0.5620$ at $a = 0.56$, against the formula's 0.5601. A fine search over a puts the maximum at $a \approx 0.5601 = 4/(4+\pi)$.
- Simulating the yearly doubling with $p = 4/(4+\pi)$ gave a mean tax after 3 years of about 3.805, against $\left(\frac{8+\pi}{4+\pi}\right)^3 \approx 3.797$.

Common slips

- Subtracting the "covers an intersection" area from the "no line" area, or counting it twice. The two events are disjoint, and both are removed from 1.
- Using the full disc πa^2 at each corner and forgetting that only a quarter of it lies in the square (or, equivalently, that each corner disc is shared by four squares).
- Writing the expected tax as $(2p)^n$, forgetting that an undoubled year leaves the tax unchanged.

Beyond STEP

This is a cousin of Buffon's needle problem: a randomly placed object is reduced to a random point in a fundamental region, and probabilities become areas. That idea is the start of *geometric probability* and integral geometry, which gives surprising results such as estimating π by throwing needles. The final part uses the fact that expectations of independent random variables multiply, which is the basis of multiplicative growth models in finance and population biology. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.